🧰 Pre-Workshop Setup Guide

Installing Python 3, Visual Studio Code (VS Code) and Github
(For both Windows and macOS users)

This guide will help you get ready for our Python for Administrative Staff Workshop. Please follow these steps before the training session so you're ready to start coding right away!

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🧩 PART 1: Installation of Python 3

🪟 For Windows Users

Step 1: Download Python 3

1. Go to https://www.python.org/downloads/
2. Click the "Download Python 3.x.x" button (the latest version).

Step 2: Run the Installer

1. Open the downloaded file (example: python-3.12.x.exe).
2. Check the box: "Add Python to PATH" before clicking Install Now.
3. Wait for the installation to complete.
4. Click Close when done.

Step 3: Verify Installation

1. Open the ***cmd*** Application. Open the Start Menu → type “cmd” → press Enter.
2. Type: python --version
3. You should see something like: Python 3.12.x

✅ If you see this, Python is successfully installed!

🍎 For macOS Users

Step 1: Check if Python is Already Installed

1. Open Terminal (press ⌘ + Space → type “Terminal”).
2. Type: python3 --version
3. If you see something like Python 3.x.x, you already have Python 3 installed.

Step 2: Install Python

1. Go to https://www.python.org/downloads/
2. Click the “Download Python 3.x.x” button.
3. Open the downloaded .pkg file and follow the instructions.

Step 3: Verify Installation

1. Open Terminal again.
2. Type: python3 --version
3. You should see something like Python 3.12.x

✅ Done! Python is ready to use.

🧩 PART 2: Install Visual Studio Code (VS Code)

(These steps are the same for both Windows and Mac.)

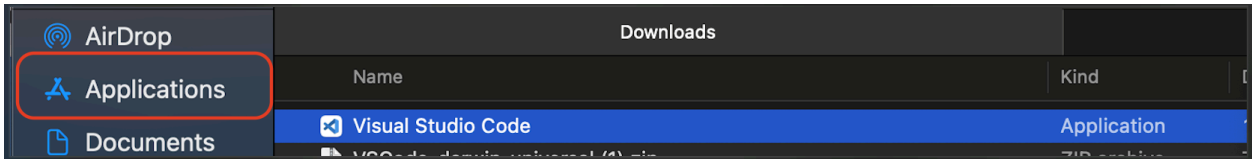
Step 1: Download VS Code

1. Go to https://code.visualstudio.com/
2. Click Download for Windows or Download for macOS.

Step 2: Install VS Code

• For Windows users: Run the .exe installer and click Next through the defaults.

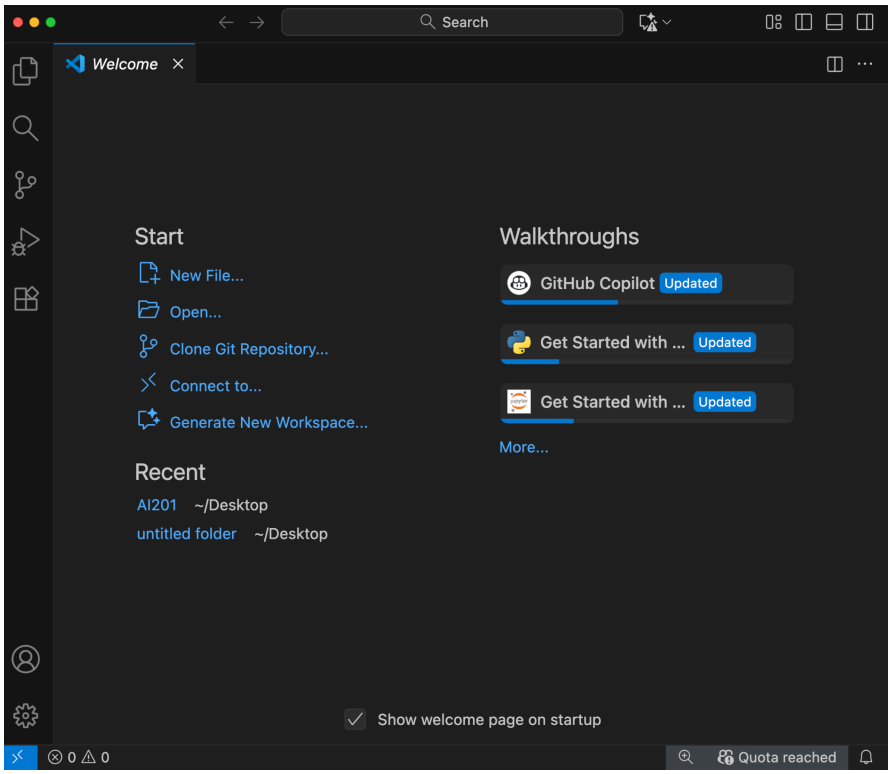
• For Mac users: Open the VSCode installer .zip file from your Downloads folder. The application will be unzipped and installed in your Downloads folder. Drag the Visual Studio Code into your Applications folder.



Step 3: Open VS Code

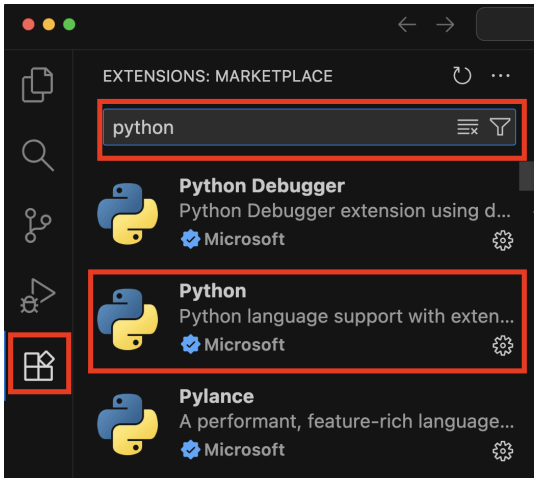
Locate and open the Visual Studio Code Application (from your Applications folder).

You should see the Welcome screen.

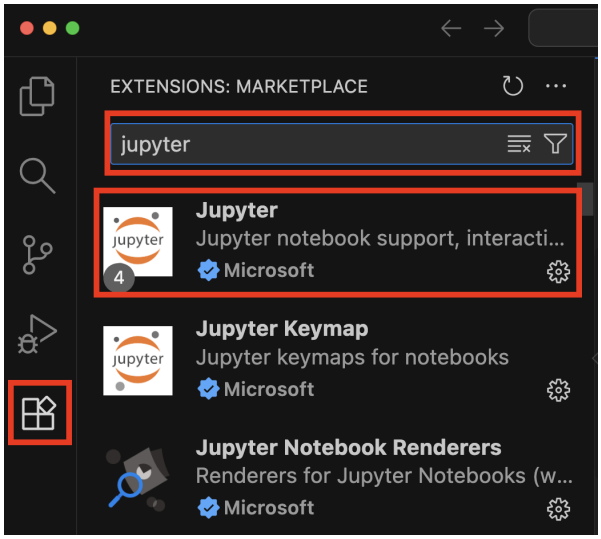


🧩 PART 3: Add the Python and Jupyter Extensions in VS Code

1. In VS Code, click on the Extensions icon (left sidebar — looks like four small squares).
2. In the search box, type Python.
3. Click Install on the official extension by Microsoft.



4. Next, search for 'Jupyter' and install the official Jupyter extension by Microsoft.



5. Restart VS Code after installation.

✅ You can now open .py and .ipynb files and run Python code directly inside VS Code.

🧩 PART 4: Enable Jupyter Notebook in VS Code

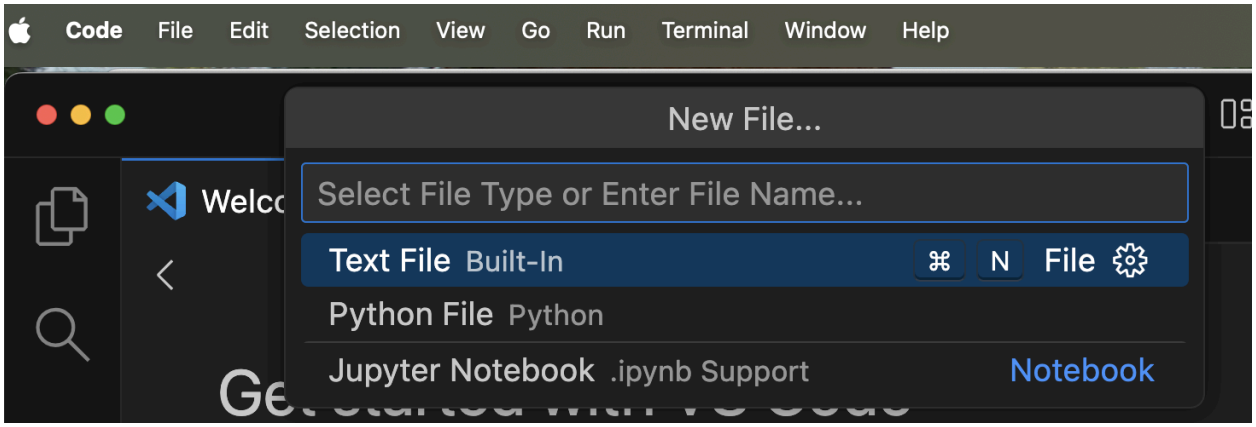
Step 1: Open VS Code and click on the Extensions icon (four squares on the sidebar).

Step 2: Make sure both 'Python' and 'Jupyter' extensions are installed and enabled.

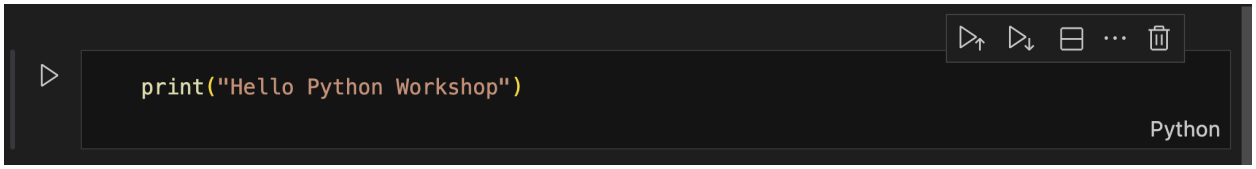
🧩 PART 5: Test Your Setup

Step 1: Open VS Code.

Step 2: Create a Python Notebook file. Click File -> New File… and choose Jupyter Notebook in the drop down of file type



Step 3: Type the following code in the first cell



Step 4: Click the Play button symbol to the left of the cell, then select the ‘default’ Python kernel to run the code.

✅ If you see 'Hello Python Workshop” printed below the cell, your setup works!

Prepare your laptop with these tools ready for the workshop. You’ll be writing your first Python + Copilot program in Week 1!

🧩 PART 6: Setting Up Git and GitHub

Git and GitHub help you save versions of your code and collaborate easily. We'll use them during the workshop to manage your exercises and share files.

💾 Step 1: Install Git

- For Windows Users : Go to https://git-scm.com/download/win and run the installer using the default options.

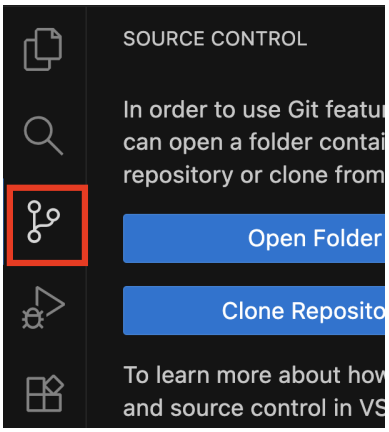
- For macOS Users : Open Terminal and type: brew install git (if Homebrew is installed), or refer to https://git-scm.com/download/mac.
- Verify installation by typing: git --version in the command prompt or terminal.

🌐 Step 2: Create a GitHub Account

1. Go to https://github.com and click 'Sign up'.
2. Follow the instructions to create your account.
3. Verify your email address when prompted.

🔗 Step 3: Connect Git with GitHub

1. Open VS Code and go to the Source Control tab (icon with a branch sym

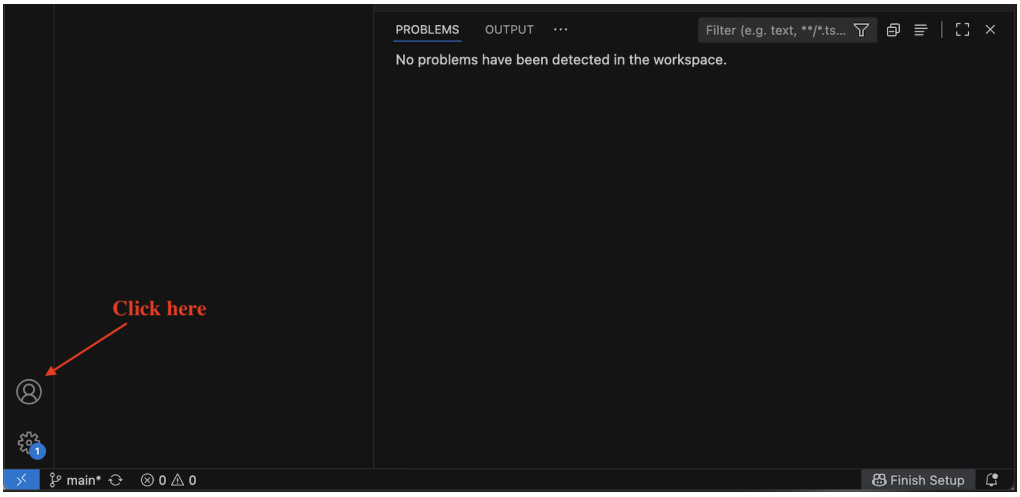


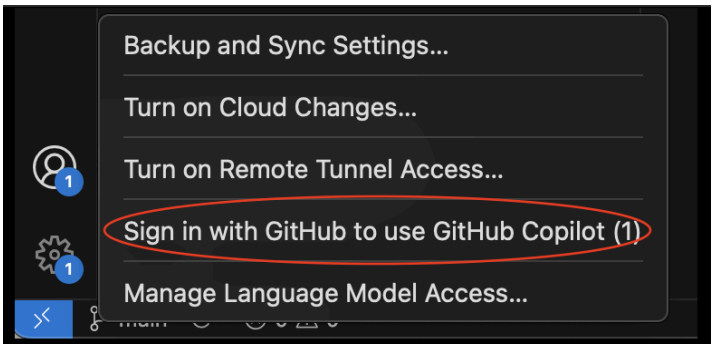
2. If prompted, click 'Install Git' or 'Sign in to GitHub'.

3. Follow the login prompts to connect your GitHub account with VS Code.

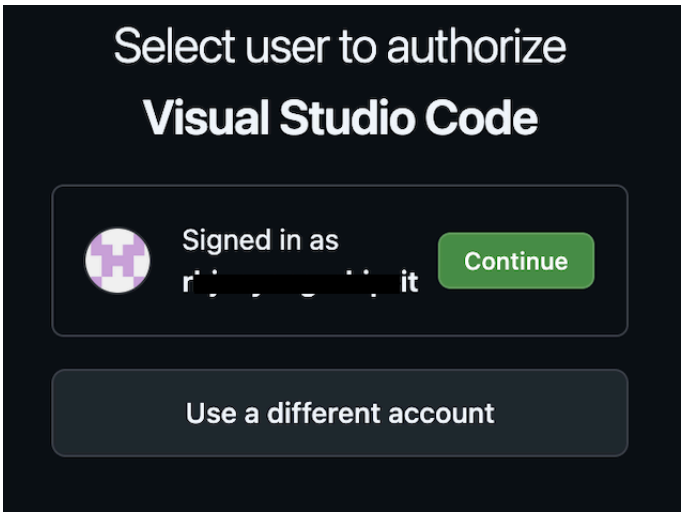
   Signing in to Github

   Click the ‘Account’ icon at the bottom of the VS Code window.

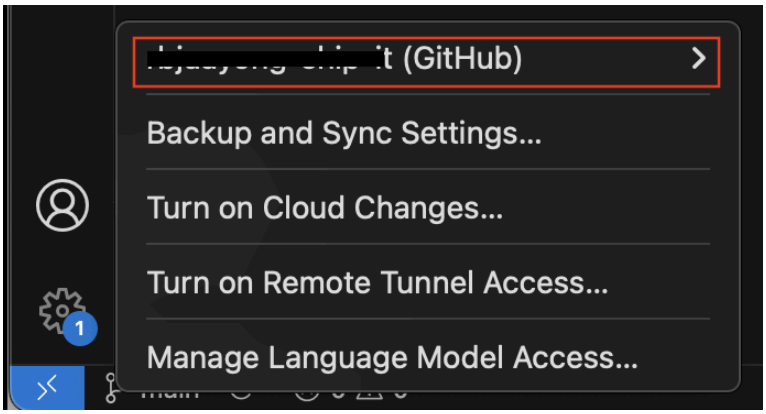




Upon clicking ‘Sign in’, a Github page for signing in your account will open in your web browser.



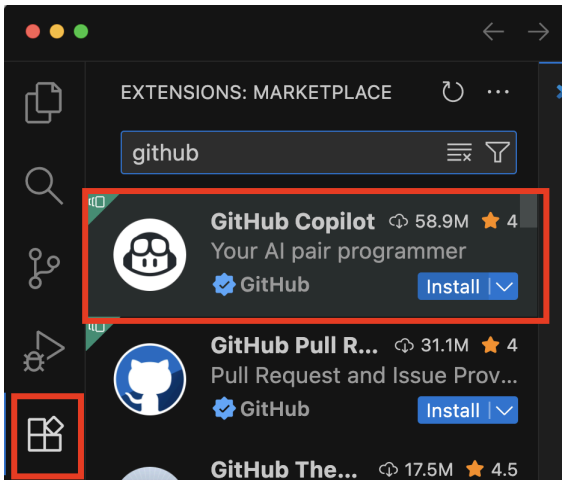
4. Once connected, you’ll see your GitHub username at the bottom-left corner of VS Code.



🧩 PART 7: Setting Up Copilot in VSCode

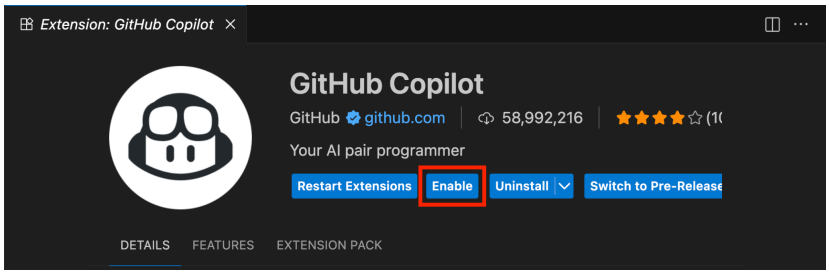
Step 1: Open VS Code and click on the Extensions icon (four squares on the sidebar).

Step 2: Search for **GitHub Copilot** and install it.



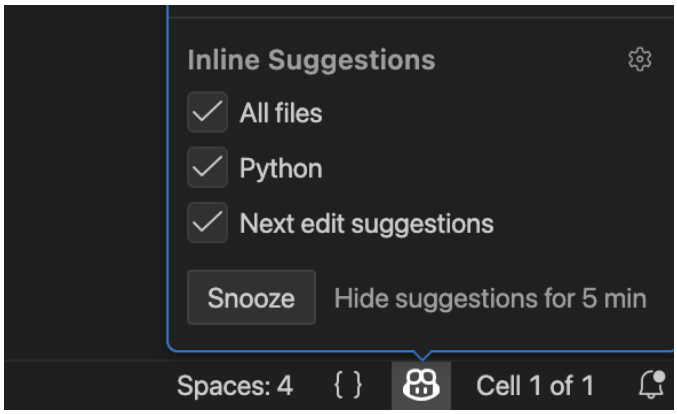
Step 3: When prompted, log in to your **GitHub account** (you may be asked to authorize Copilot),

Step 4: After installation, enable the GitHub Copilot by clicking the 'Enable' button



Step 5: Verify that Copilot is active by checking for the Github Copilot icon in the status bar located at the bottom rightmost of the window.

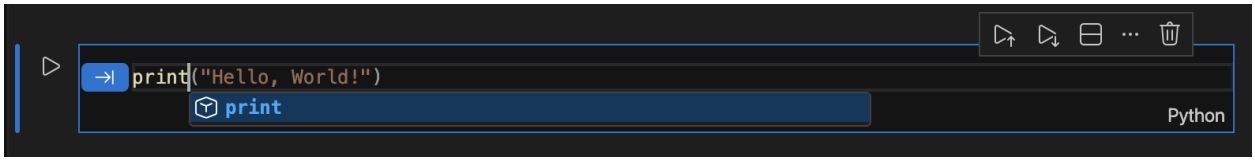




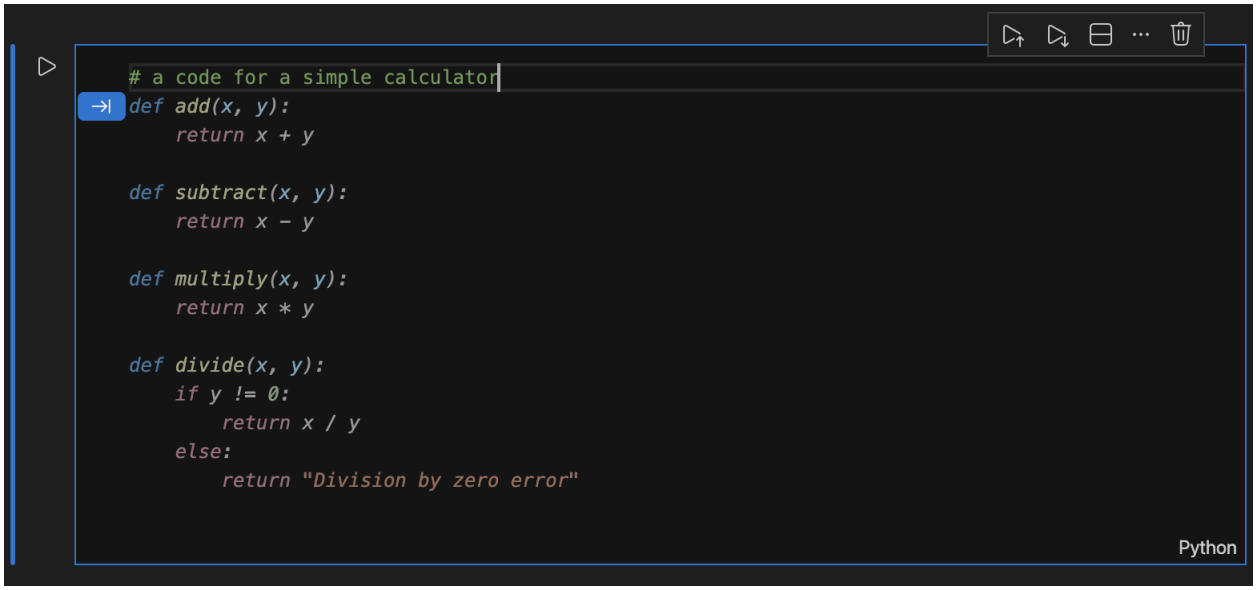
Step 6: Copilot is used in VSCode for suggesting code or code edits. To test your copilot, type in some Python code -- you'll see Copilot suggestions in gray text.

1. Create a Python Notebook file. Click File -> New File… and choose Jupyter Notebook in the drop down of file type
2. Start typing some Python code — you'll see Copilot suggestions in gray text.

   The illustration below shows a suggestion upon typing *print.*



Here is another Copilot suggestion upon typing: # a code for a simple calculator




```
# a code for a simple calculator
def add(x, y):
    return x + y

def subtract(x, y):
    return x - y

def multiply(x, y):
    return x * y

def divide(x, y):
    if y != 0:
        return x / y
    else:
        return "Division by zero error"
```

- Press **Tab** to accept a suggestion.
- Press **Esc** to ignore it.

For more information regarding setting up Github Copilot in VS Code, you may visit: https://code.visualstudio.com/docs/copilot/setup